

Smart Door Lock System using BLYNK

Project Aim

Design and implement a Smart Door Lock System using ESP32, Servo Motor, and Blynk IoT for remote door locking and unlocking through a mobile application..

Project Overview

This project uses an ESP32 connected to the Blynk Cloud via Wi-Fi. A Button Widget in the Blynk mobile app controls a servo motor acting as a door lock. When the button is pressed, the servo rotates to 90° to unlock the door and updates the status on the Blynk dashboard. When released, the servo returns to 0° to lock the door. The system provides real-time monitoring and control over the internet.

Components Used

- ESP32 Development Board
- Servo Motor
- Jumper Wires
- Blynk IoT Account

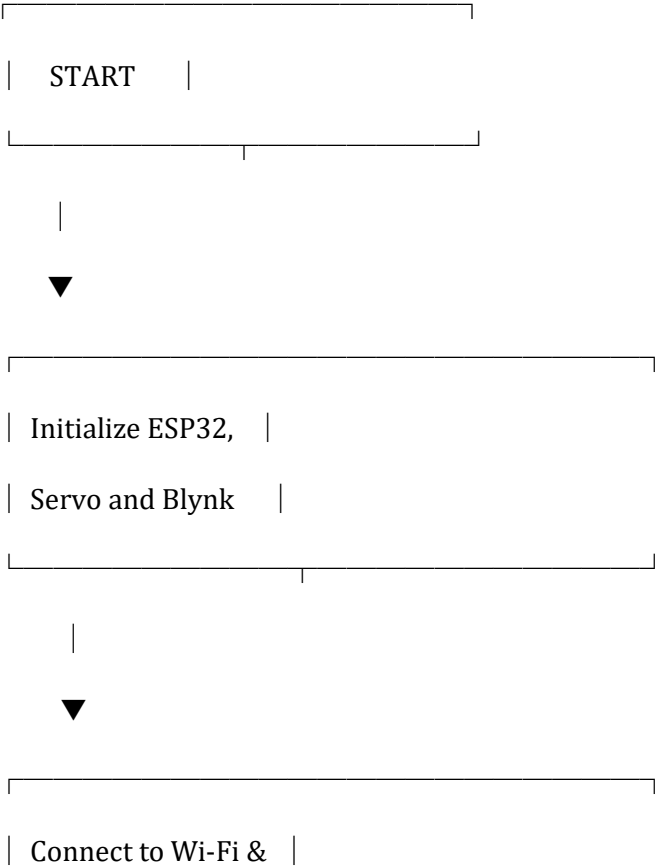
Construction and Wiring Details

ESP32 Pin	Servo Motor
GPIO 18	Signal (Orange/Yellow)
5V	VCC (Red)
GND	GND (Brown/Black)

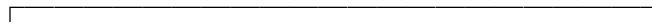
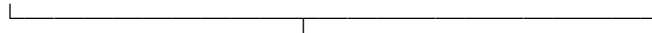
Blynk Connection

Widget	Virtual Pin	Purpose
Button	V0	Lock/Unlock Control
Label/Value Display	V1	Door Status Display

Code Structure / Program Flow

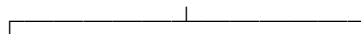
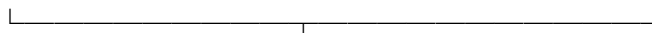


| Blynk Cloud |

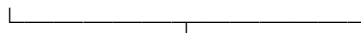


| Wait for Button |

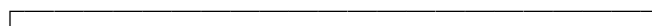
| Command from Blynk |



| Button ON? |

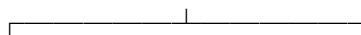
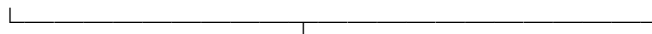


| Yes

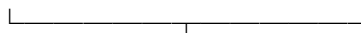


| Rotate Servo to 90° |

| Display "Unlocked" |

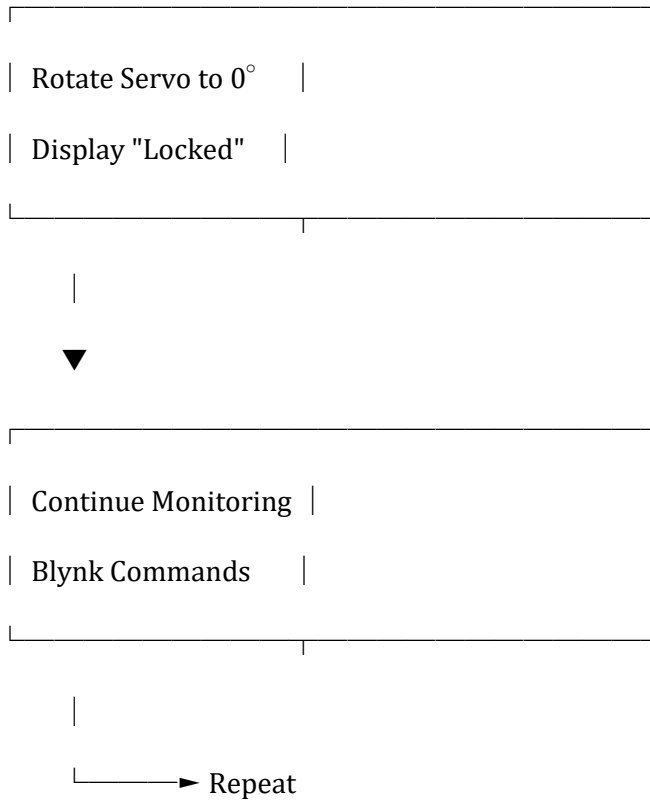


| Button OFF |



| Yes





Source Code

Diagram.jetson

```
{  
  "version": 1,  
  "author": "Aman Abdulla",  
  "editor": "wokwi",  
  "parts": [  
    { "type": "board-esp32-devkit-c-v4", "id": "esp", "top": 0, "left": 0, "attrs": {} },  
    {  
      "type": "wokwi-servo",  
      "id": "servo1",  
      "top": -77.8,  
      "left": -180.6,
```

```

    "rotate": 270,

    "attrs": {}

}

],

"connections": [

    [ "esp:TX", "$serialMonitor:RX", "", [] ],

    [ "esp:RX", "$serialMonitor:TX", "", [] ],

    [ "servo1:GND", "esp:GND.1", "black", [ "v29.8", "h69", "v57.6", "h-9.6" ] ],

    [ "servo1:V+", "esp:5V", "green", [ "v0" ] ],

    [ "esp:18", "servo1:PWM", "green", [ "h33.64", "v-124.8", "h-192", "v96", "h-28.8" ] ]

],

"dependencies": {}

}

```

Esp32 Code

```

#define BLYNK_TEMPLATE_ID "TMPL37V_b0h9-"

#define BLYNK_TEMPLATE_NAME "Smart Door Lock System"

#define BLYNK_AUTH_TOKEN "S_o7MsD58RGEyyYY8wVwgrylBKhdvQG1"


#include <WiFi.h>

#include <WiFiClient.h>

#include <BlynkSimpleEsp32.h>

#include <ESP32Servo.h>


char ssid[] = "Wokwi-GUEST";

```

```
char pass[] = "";
```

```
Servo doorServo;
```

```
#define SERVO_PIN 18
```

```
BLYNK_WRITE(V0)
```

```
{
```

```
  int value = param.asInt();
```

```
  if (value == 1)
```

```
  {
```

```
    doorServo.write(90);    // Unlock
```

```
    Blynk.virtualWrite(V1, "Door Unlocked");
```

```
    Serial.println("Door Unlocked");
```

```
  }
```

```
  else
```

```
  {
```

```
    doorServo.write(0);    // Lock
```

```
    Blynk.virtualWrite(V1, "Door Locked");
```

```
    Serial.println("Door Locked");
```

```
  }
```

```
}
```

```
void setup()
```

```
{
```

```
Serial.begin(115200);
```

```
doorServo.attach(SERVO_PIN);
```

```
doorServo.write(0); // Initially locked
```

```
Blynk.begin(BLYNK_AUTH_TOKEN, ssid, pass);
```

```
Blynk.virtualWrite(V1, "Door Locked");
```

```
}
```

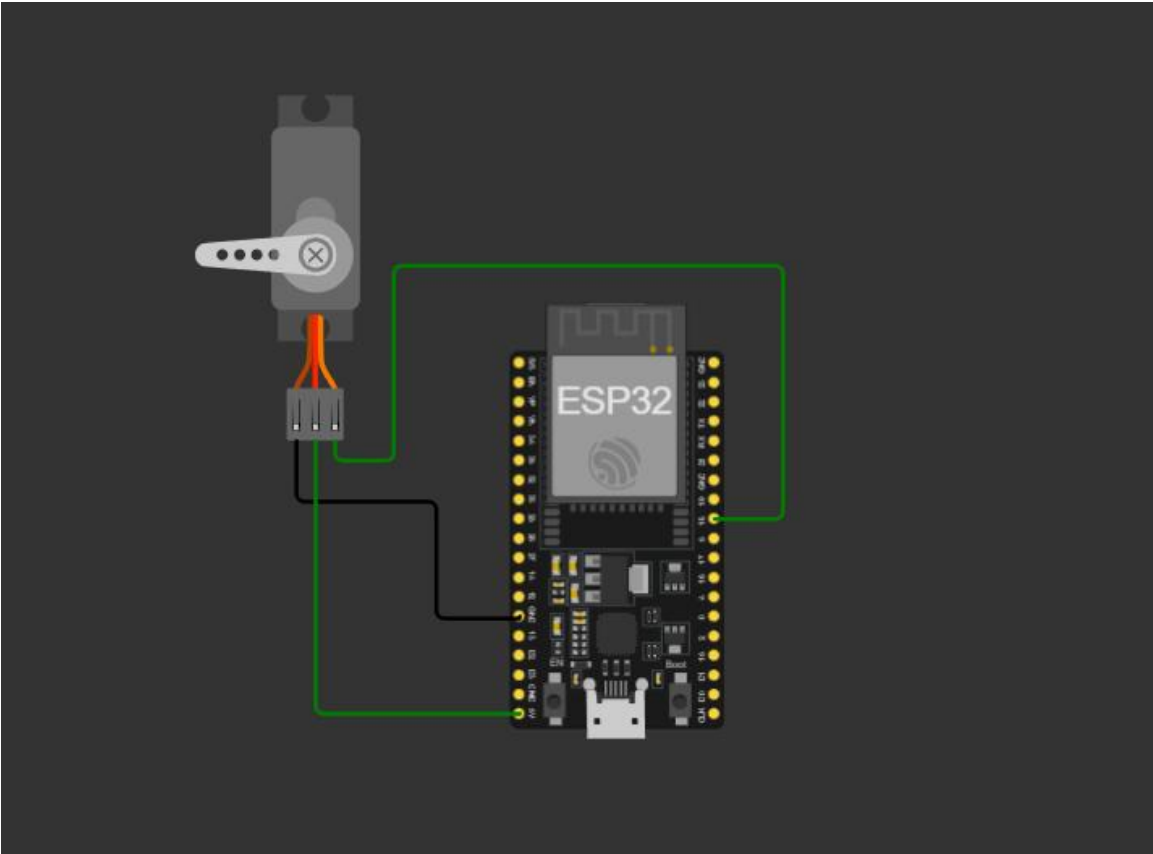
```
void loop()
```

```
{
```

```
  Blynk.run();
```

```
}
```

Project Photograph





Smart Door Lock System Online

Auth Token:.....vQG1

Aman

My organization - 6357ZA

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🔔

🔧

📄

⋮

Edit

Live

1h

6h

1d

1w

1mo

3mo

6mo

1y

📅

Door Control

☐

Door status

Door Locked

Links

Google drive :

https://drive.google.com/drive/folders/1rh0I_s0CsxwmNnfl6H_klPUgeCI6f87S?usp=sharing

Wokwi : <https://wokwi.com/projects/466475529334031361>

Git hub repo : <https://github.com/AmanAZ5950/CIRCUITRON.git>